

Ruichen Xu

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SUMMARY

Ph.D. candidate in Computational Applied Mathematics developing **machine learning and deep learning models** with reproducible training and evaluation pipelines. Research spans scientific representation learning, generative modeling and optimization, with a **560,000-record benchmark** and coauthored ICLR, TMLR and IJCNN papers.

EDUCATION

Stony Brook University

Ph.D., Computational Applied Mathematics | GPA: 4.00/4.00 | Expected 2027

Aug 2022 – Present

Stony Brook, NY

Advisor: Professor Yuefan Deng

New York University – Courant Institute

M.S., Mathematics | GPA: 4.00/4.00

Sep 2020 – May 2022

New York, NY

University of California, Davis

M.S., Statistics | GPA: 3.94/4.00

Sep 2019 – Jun 2020

Davis, CA

Beijing University of Chemical Technology

Bachelor's degree, Financial Mathematics | Minor: Commercial Management

Sep 2015 – Jun 2019

Beijing, China

PROFESSIONAL EXPERIENCE

Stony Brook University

Aug 2022 – Present

Doctoral Research — Selected Projects

JEPA for Science

Jan 2026 – Present

- Developed **joint-embedding representation learning** for scientific graphs and sparse PDE observations, supporting classification, regression and physical field recovery. Built pretraining, adaptation and evaluation workflows that improved performance on multiple graph tasks and enabled deterministic single-pass inference from partial PDE observations.

Related publications: *HP-JEPA* [15]; *JENO: Full-Field Latent Prediction* [16]; *SCOPE* [13].

PDE-OBS: Controlled Evaluation Across Observation Patterns

2025 – Present

- Built a reproducible pipeline and a **560,000-record resource** spanning seven PDE families to evaluate models under changing observation patterns. Standardized inputs, prediction targets and data splits for fair comparisons, revealing systematic accuracy degradation under observation shifts and evaluating mixed-pattern training to address these failures. [\[Code\]](#)

Related publications: *PDE-OBS: Controlled Evaluation Across Observation Patterns* (preprint, 2026) [14].

ORACLE: LLM-Guided Molecular Optimization

2025 – Present

- Built a configurable molecular optimization workflow combining **LLM-proposed edits**, simulated annealing, validity checks and candidate scoring. Integrated docking, drug-likeness, synthetic-accessibility and diversity objectives, comparing search operators, schedules and acceptance rules in tracked experiments to analyze multi-objective design trade-offs.

Efficient Generative Models & VLM Post-Training

2025 – 2026

- Coauthored research on **computational and data efficiency** in generative models and vision-language post-training, spanning 3D molecular generation and temporal video grounding. The studies demonstrated shorter molecular generation trajectories and stronger temporal grounding accuracy using fewer examples and less training time in the evaluated comparisons.

Related publications: *GAGA* (ICLR 2026) [1]; *Less Data, Better Timing* (preprint, 2026) [17].

Physics-Aware Dynamics, Diffusion & Optimization

2025

- Developed methods for recovering dynamics from position measurements, constrained diffusion sampling and adaptive optimization. Led and co-led research combining physical structure with neural modeling and reinforcement learning, contributing **first-author and co-first-author work** in Hamiltonian dynamics, diffusion inference and simulated annealing.

Related publications: *KAR-HNN* [3]; *Velocity-Inferred Hamiltonian Networks* [8]; *APOD* [11]; *RL-QESA* [12].

SPIR (Stony Brook University)

Feb – Apr 2026

Research Project Assistant

On-campus graduate RA through the Research Foundation for SUNY; Sunrise Technology, Inc. project at CEWIT.

- Collected camera images during manual driving of a miniature autonomous car and paired visual observations with recorded control actions. Trained an end-to-end **behavior cloning model** to learn a driving policy from these human demonstrations.
- Trained driving policies with **on-policy reinforcement learning** for a miniature autonomous car, with primary responsibility for **reward-function design and refinement** using visual feedback to guide policy learning and autonomous driving behavior.

SELECTED PUBLICATIONS

Name in bold; * denotes co-first author. Full list: [Homepage](#) / [Google Scholar](#)

Conference & Journal Papers

- [1] **GAGA: Gaussianity-Aware Gaussian Approximation for Efficient 3D Molecular Generation.** J. Qu, W. Gao, **Ruichen Xu**, Y. Liu. *ICLR 2026*.
- [2] **Discretization-invariance? On the Discretization Mismatch Errors in Neural Operators.** W. Gao, **Ruichen Xu**, Y. Deng, Y. Liu. *ICLR 2025*.
- [3] **Kolmogorov-Arnold Representation for Symplectic Learning: Advancing Hamiltonian Neural Networks.** Z. Wu, **Ruichen Xu***, et al. *IJCNN 2025*. Co-first author; project lead.
- [4] **Dynamic Schwartz-Fourier Neural Operator for Enhanced Expressive Power.** W. Gao, J. Luo, **Ruichen Xu**, Y. Liu. *TMLR 2025*.
- [5] **Coordinate Transform Fourier Neural Operators for Symmetries in Physical Modeling.** W. Gao, **Ruichen Xu**, H. Wang, Y. Liu. *TMLR 2024*.
- [6] **Towards Principled Test-Time Adaptation for Time Series Forecasting.** H. Wang, **Ruichen Xu**, et al. *TMLR 2026*; *accepted with minor revision*.
- [7] **Multistep Generative Backmapping of Coarse-Grained Structures.** G. Kementzidis, E. Wong, J. Nicholson, **Ruichen Xu**, Y. Deng. *Computer Physics Communications* 327, 110286 (2026).
- [8] **Velocity-Inferred Hamiltonian Networks: Symplectic Dynamics from Position-Only Observations.** **Ruichen Xu**, C. Yu, et al. *NYSDS 2025, SIAM Proceedings*.
- [9] **Boundary-Informed Method of Lines for Physics-Informed Neural Networks.** M. Cederholm, S. Wang, H. Wang, **Ruichen Xu**, Y. Deng. *NYSDS 2025, SIAM Proceedings*. Co-corresponding author.
- [10] **Physics-Informed Active Learning via Functional Simulated Annealing for Neural Operator.** A. Ding, S. Wang, H. Wang, **Ruichen Xu**, Y. Deng. *NYSDS 2025, SIAM Proceedings*. Co-corresponding author.

Workshop Papers

- [11] **APOD: Adaptive PDE-Observation Diffusion for Physics-Constrained Sampling.** **Ruichen Xu**, H. Wang, et al. *ICML 2025 Workshop on Assessing World Models (poster)*.
- [12] **RL-QESA: Reinforcement-Learning Quasi-Equilibrium Simulated Annealing.** **Ruichen Xu**, K. Li, et al. *2nd AI for Math Workshop at ICML 2025 (poster)*.

Preprints & Papers Under Review

- [13] **SCOPE: Observation-Conditioned Full-Target Prediction for Sparse PDE Inference.** **Ruichen Xu**, et al. *arXiv:2609.36527, 2026*; *under review at ICLR 2027*.
- [14] **PDE-OBS: Controlled Evaluation Across Observation Patterns.** **Ruichen Xu**, et al. *arXiv:2609.36521, 2026*.
- [15] **HP-JEPA: Hierarchical Partitioning for Multi-Resolution Graph Joint-Embedding Predictive Learning.** **Ruichen Xu**, et al. *arXiv:2608.00491, 2026*.
- [16] **JENO: Full-Field Latent Prediction for Sparse Inverse PDE Inference.** **Ruichen Xu**, et al. *2026*; *under review*.
- [17] **Less Data, Better Timing: Student-Curriculum Coupling for VLM On-Policy Distillation in Temporal Video Grounding.** J. Qiu, Y. Kim, **Ruichen Xu**, et al. *arXiv:2609.40055, 2026*.
- [18] **The Impact of Move Schemes on Simulated Annealing Performance.** **Ruichen Xu**, H. Wang, Y. Deng. *arXiv:2504.17949, 2025*.

MENTORING & TEACHING

AI3 REU, Stony Brook University | Research Mentor

Jun – Jul 2026

Mentored AI4Science project scoping, experiment design and implementation; guided reproducible workflows and research presentations.

Stony Brook University | Instructor & Teaching Assistant

Aug 2022 – Jan 2026

Taught Statistical Laboratory, Numerical Analysis and Differential Equations; designed lectures and assessments, and led recitations, office hours and grading.

NYU Courant | Recitation Leader & Grader

2020 – 2022

Led recitations in Mathematics for Economics II and Probability, Statistics & Decision Making; graded Analysis and Special Topics coursework.

RECOGNITION, ACADEMIC SERVICE & TALKS

ICML 2026 Gold Reviewer. Conference service: ICLR (2026/2027), NeurIPS (2026), IJCNN (2025/2026), KDD (2026/2027); AAAI 2027 Program Committee.

Journal / workshop reviewing: TMLR, IEEE TNNLS, Neurocomputing; AI for Math at ICML 2025.

IACS presentations: AI for PDEs (2025); Diffusion4PDE (2025); Active Learning for Neural Operators (2024).

TECHNICAL SKILLS

Programming & Computing: Python, PyTorch; GPU/HPC experiment workflows, SLURM.

Modeling: Representation learning, neural operators, graph learning, diffusion models, reinforcement learning, behavior cloning, on-policy RL.

Evaluation & Optimization: Benchmarking, ablations, distribution-shift evaluation, simulated annealing, molecular search.